

HANDHELD EMBEDDED CALCULATOR

Adriano Zagar

Tech Stack: C++, Arduino, ESP32, KiCad, SOLIDWORKS

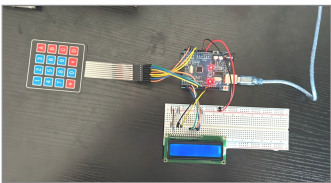
OBJECTIVE: To design and deliver a fully functional, production-ready handheld consumer electronic device from concept to prototype, successfully integrating custom C++ firmware, hardware, and mechanical enclosure design while optimizing for manufacturing cost and spatial efficiency.



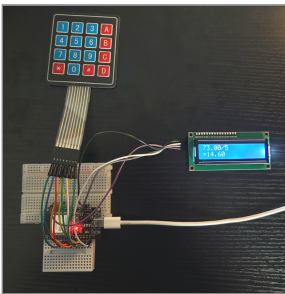
Final Assembled Handheld Calculator

DEVELOPMENT WORKFLOW

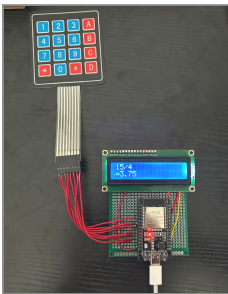
- **1. Breadboard Prototype:** Validated core computational logic and C++ firmware using an Arduino Uno backend.
- **2. Architecture Migration:** Re-engineered the system around an ESP32 and 4-pin LCD to minimize wiring complexity and footprint.
- **3. Hardware Validation:** Built and tested an interim perf-board prototype to verify signal and power integrity.
- **4. Custom PCB Design:** Engineered a cost-optimized 2-layer schematic and PCB in KiCad.
- **5. Mechanical Enclosure:** Designed a compact snap-fit housing in SOLIDWORKS.



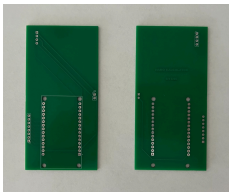
Arduino Uno Breadboard



ESP32 Breadboard



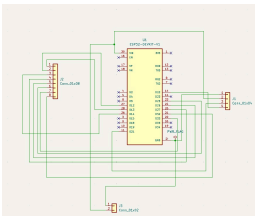
Perf-Board



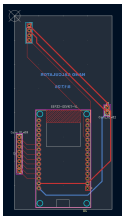
Custom PCB

ENGINEERING CHALLENGES & SOLUTIONS

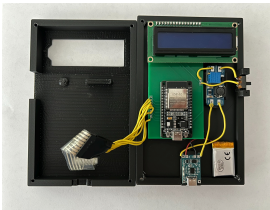
- **Cross-Domain Integration:** Unified custom electronics, C++ firmware, and a 3D-printed enclosure into a functional handheld unit.
- **Spatial Constraints:** Optimized component placement and low-profile parts for the handheld form factor.
- **Rechargeable Power System:** Integrated an MT3608 step-up module, 3.7 V rechargeable battery, and TP4056 charger module.
- **Design for Manufacturability (DFM):** Designed for efficient assembly, durability, and cost-effective sourcing.



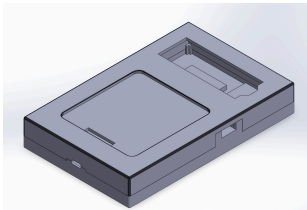
Schematic (KiCad)



PCB Layout (KiCad)



Internal Electronics Assembly



SOLIDWORKS Enclosure CAD